

Stopping times

- ▶ Let $X_1, X_2, \dots$ be a sequence of independent random variables.
- ▶ An integer valued positive random variable N is said to be a stopping time for this sequence if the event $\{N = n\}$ is independent of $X_{n+1}, X_{n+2}, \dots$ for $n = 1, 2, \dots$
- ▶ N is not independent of the entire sequence $\{X_i\}$.
- ▶ Think as if we are seeing X'_n s one at a time and stop after a stopping criteria is met.
- ▶ So if we stop after seeing $X_1, X_2, \dots, X_n$, then $N = n$.
- ▶ Suppose $P(X_n = 1) = P(X_n = -1) = 0.5$. Then $N = \min\{n : X_1 + \dots + X_n = 1\}$ is a stopping time.
- ▶ Stop one roll before you see 6. Is this a stopping time ?No.

Stopping times for Renewal process

- ▶ Is $N(t)$ a stopping time for the sequence of interarrivals X_i ?
- ▶ Suppose $N(t) = n$, i.e., by time t there have been only n arrivals. Then what we know is that $S_n \leq t$ and that $S_{n+1} > t$.
- ▶ Therefore $N(t) = n$ depends on X_{n+1} . For it to be a stopping time, it should have been independent of X_{n+1} .
- ▶ Therefore $N(t)$ is not a stopping time.
- ▶ However $N(t) + 1$ is a stopping time. This is because $N(t) + 1 = n$ implies $N(t) = n - 1$ for which $S_{n-1} \leq t$ and that $S_n > t$.
- ▶ $N(t) + 1 = n$ depends on $X_1, \dots, X_n$ and is independent of $X_{n+1}, X_{n+2}, \dots$.

Wald's Equation

Theorem

If $X_1, X_2, \dots$, are independent and identically distributed random variables having finite expectations, and if N is a stopping time for $X_1, X_2, \dots$ such that $E[N] < \infty$, then

$$E\left[\sum_{i=1}^N X_i\right] = ENEX$$

Proof on board. Also Refer Sheldon Ross, Thm 3.3.2.

Corollary

$$E[S_{N(t)+1}] = E[X](m(t) + 1)$$

RECAP

- ▶ Renewal Process $N(t)$ with arbitrary interarrival times
- ▶ Renewal equation: $m(t) = \int_0^t (1 + m(t-x))dF(x)$
- ▶ Stopping times and Wald's Equation:

$$E\left[\sum_{i=1}^N X_i\right] = ENEX$$

- ▶ Stopping times for renewal process is $N(t) + 1$
- ▶ $E[S_{N(t)+1}] = E[X](m(t) + 1)$

Time average versus Ensemble average

$$\bar{X}^{time-avg} = \lim_{t \rightarrow \infty} \frac{\int_0^t X(u, \omega) du}{t}$$

$$\bar{X}^{ensemble} = \lim_{t \rightarrow \infty} E(X(t))$$

For an ergodic process, $\bar{X}^{time-avg} = \bar{X}^{ensemble}$

- ▶ Consider a Markov coin (with unknown transition probabilities) and given a budget of 10,00,000 (10 lakh) tosses, how will you find the stationary probability of head?
- ▶ Exhaust all at once (time average)
- ▶ Perform 100 runs each of length 10000 and average across the last toss in each run! (ensemble average)

Renewal theorem

Lemma

- ▶ With probability 1, $\frac{N(t)}{t} \rightarrow \frac{1}{E[X_1]}$ as $t \rightarrow \infty$.
(Proof hint:- $S_{N(t)} \leq t \leq S_{N(t)+1}$)
- ▶ $\frac{m(t)}{t} \rightarrow \frac{1}{E[X_1]}$ as $t \rightarrow \infty$.

See Sheldon Ross (Stochastic Processes, 2nd edition) Proposition 3.3.1 and Thm 3.3.4 for proof.

NOTE: $S_{N(t)+1} > t$. Taking Expectations on both sides, invoking Wald's lemma, and rearranging gives us $\liminf_{t \rightarrow \infty} \frac{m(t)}{t} \geq \frac{1}{E[X_1]}$

Renewal Reward theorem

- ▶ Consider a renewal process with interarrival times $X_i, i = 1, 2, \dots$. Suppose a random reward Y_i is earned at the time of the i th arrival. While Y_i may depend on X_i , the pairs (X_i, Y_i) are independent and identically distributed.
- ▶ Let $Y(t)$ denote the total reward accrued till time t . Then $Y(t) = \sum_{i=1}^{N(t)} Y_i$.

Lemma

- ▶ With probability 1, $\frac{Y(t)}{t} \rightarrow \frac{E[Y]}{E[X]}$ as $t \rightarrow \infty$.
- ▶ $\frac{E[Y(t)]}{t} \rightarrow \frac{E[Y]}{E[X]}$ as $t \rightarrow \infty$.

See Sheldon Ross Theorem 3.6.1 for proof.

Renewal reward theorem – Application to $M/M/1/1$

On board